

Datazine Facebook Analytics

SQL-Based Audience, Engagement & Growth Analysis



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Introduction

This presentation presents the results of a SQL-based analysis of Datazine's Facebook audience and engagement metrics.

The objective of the analysis is to evaluate the effectiveness of Datazine's Facebook strategy by examining audience demographics, geographic distribution, engagement trends, and content performance patterns.

The analysis also evaluates the potential for launching an English-language edition of the magazine and provides recommendations to support future audience growth and market expansion.

Process

Data

- Facebook social media analytics data exported in CSV format
- Audience, engagement, language, geographic distribution, and post performance metrics

Data Organization

- Imported raw CSV files into SQLite tables
- Implemented the provided relational database schema
- Connected datasets using CountryCode relationships
- Validated data consistency across tables

Analysis

- Used SQL queries to identify audience and engagement trends
- Analyzed demographics, geographic distribution, posting performance, and language segmentation
- Prepared business insights and recommendations to support social media growth

Basic Summary Statistics

Basic Summary Statistics

Countries Represented:	45
Cities Represented:	46
Languages Represented:	41
Daily Global Reach:	83.8M
Avg. Daily Page Likes:	402K
Top Fan Country:	Ivory Coast 112K fans (latest reporting date)

Note: The FansPerCity dataset contained fewer countries due to higher reporting thresholds. Country-level metrics were validated using the FansPerCountry and GlobalPage tables.

Location Analysis

Top 10 Countries by Number of Fans

	Country Code	Country	Number of Fans
1	CI	Ivory Coast	112,160
2	CM	Cameroon	102,211
3	SN	Senegal	83,561
4	FR	France	73,252
5	MG	Madagascar	72,956
6	CD	Democratic Republic of the Congo	50,705
7	BF	Burkina Faso	43,500
8	ML	Mali	40,578
9	DZ	Algeria	39,093
10	GN	Guinea	36,821

Key takeaway:

- The largest fan bases are concentrated in African markets, particularly Ivory Coast, Cameroon, and Senegal.
- Ivory Coast represents the largest fan market with 112K fans.

Top 10 Countries by Penetration Ratio

	Country	Penetration Ratio	Number of Fans	Population
1	Reunion	2.41%	20,885	866,506
2	French Polynesia	1.82%	5,148	283,007
3	New Caledonia	1.79%	5,032	280,460
4	Mauritius	1.77%	24,210	1,364,283
5	Martinique	1.44%	5,427	376,480
6	Guadeloupe	1.36%	5,379	395,700
7	Gabon	1.13%	23,954	2,119,036
8	Mayotte	0.73%	1,983	270,372
9	Comoros	0.60%	4,925	821,164
10	French Guiana	0.57%	1,687	296,711

Key Takeaway:

- The data shows particularly strong fan penetration across French-speaking African and island markets.
- Reunion leads with a penetration ratio of 2.41%, indicating an exceptionally strong fan base relative to population size.

Cities with Growth Potential

	Country	City	Number of Fans	Population
1	Algeria	Bejaia	2,301	41,657,488
2	Cameroon	Ngaoundere	2,318	25,640,965
3	Madagascar	Fianarantsoa	2,366	25,683,610
4	Algeria	Tizi Ouzou	2,524	41,657,488
5	Canada	Montreal	2,887	35,881,659
6	Algeria	Oran	2,920	41,657,488
7	Ivory Coast	Bouake	3,376	24,290,000
8	Ivory Coast	Cocody	3,647	24,290,000
9	Morocco	Casablanca	3,951	34,314,130
10	Angola	Luanda	4,614	30,355,880

Key Takeaway:

- The analysis highlights cities with relatively low fan counts in countries with populations exceeding 20 million.
- Absolute fan counts alone may not fully reflect city-level fan penetration potential.

Large Cities with Lowest Fan Penetration

	City	Number of Fans	City Population	Penetration Ratio
1	Luanda	4,614	8,473,000	0.05%
2	Casablanca	3,951	3,360,000	0.12%
3	Montreal	2,887	1,763,000	0.16%
4	Fianarantsoa	2,366	901,000	0.26%
5	Oran	2,920	803,000	0.36%
6	Bouake	3,376	740,000	0.46%
7	Cocody	3,347	693,000	0.53%
8	Ngaoundere	2,318	236,000	0.98%
9	Bejaia	2,301	178,000	1.29%
10	Tizi Ouzou	2,524	135,000	1.87%

Key Takeaway:

- After adjusting for city population size, several large urban markets demonstrate relatively low fan penetration.
- Luanda and Casablanca represent the largest opportunities for fan base growth relative to their population size.

Fan Analysis

Fan Age Distribution

	Age Group	Percentage of Fans
1	25-34	35.8%
2	18-24	21.3%
3	35-44	19.4%
4	45-54	9.45%
5	65+	6.94%
6	55-64	5.02%
7	13-17	2.09%

Key Takeaway:

- Fans aged 18–44 account for approximately 77% of the total fan base, indicating strong engagement among young and middle-aged adults.
- The 25–34 age group represents the largest segment, accounting for over one-third of total fans.

Gender Distribution of Fans

	Gender	Percentage of Fans
1	F	56.41%
2	M	43.5%
3	U	0.09%

Key Takeaway:

- Female users represent the majority of the fan base, accounting for over 56% of total fans.
- The fan base shows a relatively balanced gender distribution, with females representing a slight majority.

Language Analysis

Top 5 Languages by Fans

	Language	Number of Fans	Percentage of Fans
1	French	901,668	92.88%
2	English	49,418	5.09%
3	Arabic	4,776	0.49%
4	Portuguese	4,438	0.46%
5	Spanish	3,409	0.35%

Key Takeaway:

- French is the dominant primary language among fans, representing nearly 93% of the fan base.
- English is the largest secondary language segment at 5.09%, indicating a measurable opportunity for English-language expansion.

English-Speaking Fans

	Country	Number of Fans	Buying Power
1	United States	38,909	\$200,323
2	United Kingdom	10,509	\$45,357

Key Takeaway:

- English-speaking fans represent a relatively small but high-value segment, accounting for 5.09% of the total fan base.
- The majority of English-speaking fans are concentrated in the United States and the United Kingdom.
- Combined estimated buying power of US- and UK-based English-speaking fans exceeds \$245K, indicating potential for targeted English-language campaigns.

Content & Engagement Analysis

Engagement per Day of the Week

	Day of Week	Percentage of Engagement
1	Saturday	19.73%
2	Monday	19.23%
3	Tuesday	18.67%
4	Wednesday	15.38%
5	Sunday	12.08%
6	Friday	8.59%
7	Thursday	6.32%

Key Takeaway:

- Engagement is strongest at the beginning and end of the week, with Saturday, Monday, and Tuesday showing the highest engagement levels.

Engagement per Time of Day

	Time Range	Percentage Split
1	05:00 – 08:59	33.96
2	15:00 – 18:59	17.4
3	09:00 – 11:59	15.34
4	19:00 – 21:59	13.12
5	12:00 – 14:59	12.7
6	22:00 or later	7.48

Key Takeaway:

- Early morning hours (05:00–08:59) show the highest engagement activity, accounting for nearly 34% of engaged fans.
- Additional engagement peaks occur during late morning (09:00–11:59) and afternoon-to-early-evening hours (15:00–18:59).

Posting Activity by Month

	Month	Posts	Engaged Fans	Engagements / Post
1	Jan	2	4,269	2,134
2	Feb	6	16,630	2,771
3	Mar	30	108,512	3,617
4	Apr	28	87,708	3,132
5	May	29	99,229	3,421
6	Jun	31	103,736	3,346
7	Jul	30	102,441	3,414
8	Aug	28	119,949	4,283
9	Sep	55	177,372	3,224
10	Oct	183	675,674	3,692
11	Nov	27	105,694	3,914
12	Dec	30	110,372	3,679

Key Takeaway:

- Average engagement per post remained relatively stable throughout the year, ranging between approximately 3K and 4K engaged fans.
- Overall engagement levels appear to be strongly influenced by posting frequency, suggesting that increased publishing volume generally drives higher total engagement.
- August delivered the highest engagement efficiency, generating over 4.2K engaged fans per post despite moderate posting activity.
- October likely reflects a large-scale campaign period, where exceptionally high posting volume drove peak annual engagement levels.

Month-to-Month Change in Engagement

From Month	To Month	Delta Post Clicks	Delta Engaged Fans	Delta Reach
01	02	463.64%	289.55%	313.52%
02	03	698.94%	552.51%	629.34%
03	04	-40.00%	-19.17%	-31.67%
04	05	34.75%	13.14%	21.24%
05	06	-12.78%	4.54%	0.30%
06	07	8.52%	-1.25%	1.14%
07	08	28.81%	17.09%	21.92%
08	09	38.31%	47.87%	48.09%
09	10	338.82%	280.94%	350.80%
10	11	-85.96%	-84.36%	-87.11%
11	12	2.10%	4.43%	11.58%

Key Takeaway:

- The most significant large-scale engagement expansion occurred between September and October, coinciding with a major increase in posting activity.
- The sharpest decline occurred between October and November, likely reflecting post-campaign normalization after an unusually high-volume publishing period.
- Early-year growth percentages should be interpreted cautiously due to the very low posting baseline in January and February.

Recommendations

Strategic Recommendations

1. Optimize Publishing Schedule

- Prioritize publishing on Saturdays, Mondays, and Tuesdays, which delivered the highest engagement.
- Focus posting activity between 05:00–08:59, when engagement peaks.

2. Increase Posting Volume Strategically

- Engagement appears closely linked to posting frequency.
- Increasing publishing volume could significantly expand total reach and interactions.

3. Expand English-Language Content

- English is the second-largest language segment, concentrated in the US and UK.
- Estimated combined annual buying power exceeds \$245K.
- Consider launching an English-language version of the magazine to expand reach and monetization opportunities.

4. Replicate High-Performance Campaign Patterns

- August delivered the highest engagement efficiency, while October generated the largest overall reach.
- Balance posting volume and content quality to maximize engagement.